Proposition I.3

To Cut Off from the Greater of Two Given Segments a Segment Equal to the Less

CGJteamLab

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1 Introduction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Classically, this construction is obtained by combining Proposition I.2 with an additional circle construction. A segment equal to the shorter one is first laid off from an endpoint of the greater segment, and a circle is then used to determine the required point on the greater segment.

In the present reconstruction over Hilbert geometry, the logical structure is substantially different. The theory developed in Book Zero already contains an abstract notion of one segment being less than another. This notion is defined by the existence of a point between the endpoints of the greater segment such that the corresponding subsegment is congruent to the smaller one.

Consequently, the existential content of Euclid's Proposition I.3 is already embodied in the definition of segment inequality. The reconstruction therefore consists not in repeating Euclid's geometric construction, but in exposing the witness contained in the definition of `HilbertSegmentLess`.

This proposition provides the first clear example where a classical Euclidean construction becomes an immediate consequence of the abstract language developed in Book Zero.

2 The Classical Configuration

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Unlike Proposition I.2, the present construction does not introduce a new geometric technique. Instead, it combines Proposition I.2 with one additional circle construction.

The construction proceeds as follows.

2.1 Final Configuration

The objective is to determine a point E on the greater segment AB such that

$$AE \cong CD.$$

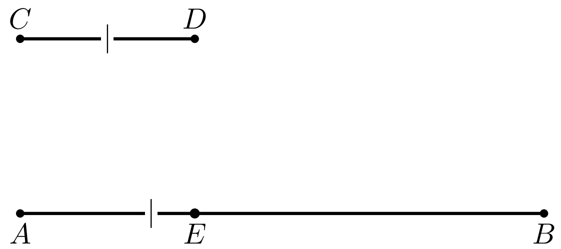

Figure 1: The final configuration of Proposition I.3: $A - E - B$ and $AE \cong CD$.

The final geometric configuration is shown in Figure 1.

From the construction we obtain

$$AE \cong CD.$$

The remaining task is to justify that the point E lies between the endpoints of the greater segment.

2.2 Construction Algorithm

The construction uses Proposition I.2 as an already established construction primitive.

2.2.1 Step 1. Given Segments

Let AB be the greater segment and CD the given shorter segment.

2.2.2 Step 2. Apply Proposition I.2

By Proposition I.2, construct a segment AF congruent to the given segment CD.

Thus

$$AF \cong CD.$$

2.2.3 Step 3. Draw the Circle Centered at A

Draw the circle centered at A passing through F.

2.2.4 Step 4. Determine the Required Point

Let E be the intersection of the circle with the greater segment.

Since

$$AE \cong AF$$

and

$$AF \cong CD,$$

it follows that

$$AE \cong CD.$$

The required segment has therefore been obtained.

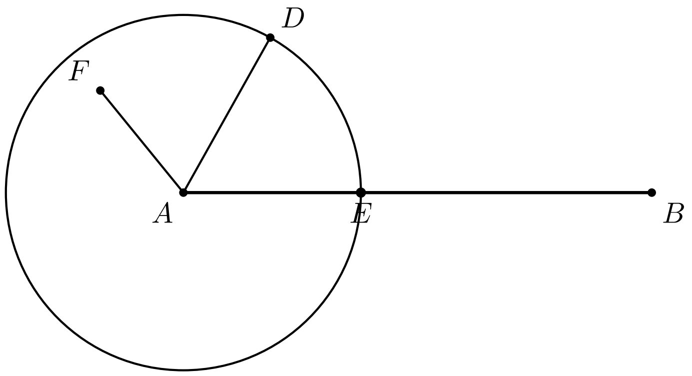

Figure 2: The classical cutting step. Proposition I.2 first provides a point D with AD equal to the given shorter segment. The circle centered at A through D meets the greater segment AB at E, so $AE = AD$. The auxiliary point F belongs to the I.2 construction and need not carry further metric markings here.

3 Classical Proof

Let AB be the greater segment and CD the shorter one. By Proposition I.2 construct a point D' such that

$$AD' \cong CD.$$

Draw the circle with center A through D'. Because $CD < AB$, this circle meets the segment AB at a point E between A and B. Every radius of the circle is congruent, so

$$AE \cong AD'.$$

Together with $AD' \cong CD$, the First Common Notion gives

$$AE \cong CD.$$

Hence the required part AE has been cut from the greater segment.

□

Diagrammatic audit. Two figures are useful. The first shows only the target relation $A{-}E{-}B$ and $AE \cong CD$. The second shows the actual classical mechanism: the transferred length from I.2 and the circle centered at A whose intersection with AB determines E. Separate snapshots for each construction instruction would add no geometric information.

Wilkins emphasizes that I.3 completes a strategy distributed over the first three propositions: I.1 supplies the equilateral-triangle construction used inside I.2, I.2 transports a length, and I.3 uses the resulting radius to cut the required part from the longer segment.

4 Proof Architecture of the Reconstruction

```
Euclid
------
CD < AB
   |
I.2: construct D with AD = CD
   |
circle centered at A through D
   |
intersection E with AB
   |
AE = AD = CD
   |
I.3


Hilbert / Lean
--------------
HilbertSegmentLess Geo C D A B
   |
unpack witness E
   |
A-E-B and CD ~= AE
   |
congruence symmetry
   |
A-E-B and AE ~= CD
   |
euclid_proposition_3
```

5 Hilbert Reconstruction

The Book Zero formulation changes the logical status of the proposition. The hypothesis

```
HilbertSegmentLess Geo C D A B
```

already contains a witness E satisfying

$$A - E - B \qquad \text{and} \qquad CD \cong AE.$$

Consequently Proposition I.3 does not invoke a segment-layoff theorem at all. It merely extracts that witness and reverses the orientation of the congruence to obtain

$$AE \cong CD.$$

This is stronger than saying that Book Zero supplies convenient infrastructure for Euclid's construction. With the present definition of strict segment comparison, the existential conclusion of I.3 is literally part of the hypothesis.

6 Lean Formalization

The complete implementation is available in the repository. The entire formal argument is worth showing because it exposes the definitional collapse of the classical construction:

```
rcases hCDAB with <E, hAEB, hCDAE>

exact <E, hAEB,
  hilbert_congruent_symmetry
    Geo C D A E hCDAE>
```

There is no hidden construction step in `Proposition03.lean`. Unpacking `hCDAB` produces exactly the required point and betweenness statement; only congruence symmetry is needed to match the orientation of the Euclidean conclusion.

7 Relationship with Earlier Propositions

7.1 Proposition I.2

Proposition I.3 is the first direct application of the segment-transfer construction established in I.2.

I.2 says that a segment congruent to a given segment can be placed at a prescribed point. I.3 adds an order constraint: when the target segment is longer than the given one, the transferred length can be realized as an initial part of that longer segment.

Thus I.3 converts free segment transport into controlled segment cutting.

7.2 Proposition I.1

Proposition I.1 enters indirectly through I.2. Euclid's segment-transfer construction uses equilateral triangles as part of the mechanism for moving a length to a new position.

By the time I.3 is reached, this construction has already been packaged as Proposition I.2 and can be used as a reusable operation.

7.3 Transition to Proposition I.4

Propositions I.1–I.3 form the initial construction block of Book I: equilateral triangle construction, segment transport, and segment cutting.

Proposition I.4 changes the character of the development. It is no longer primarily a construction theorem but a rigidity theorem for triangles: SAS.

Thus I.3 closes the first elementary segment-construction sequence and prepares the congruence theory beginning with I.4.

8 Hilbert and Book Zero Components Used

8.1 Strict Segment Comparison

The decisive object is `HilbertSegmentLess`. In the present Book Zero language, $CD < AB$ is witnessed by an interior point E of AB whose initial subsegment is congruent to CD. Thus order and congruence are already coupled in the definition.

8.2 Betweenness

The witness contains the positional fact

$$A - E - B.$$

This is the formal counterpart of the classical requirement that the circle meet the greater segment itself, not merely its supporting line.

8.3 Congruence Symmetry

The witness supplies $CD \cong AE$, while the proposition is stated with $AE \cong CD$. The only theorem used after unpacking the witness is segment-congruence symmetry.

9 Formalization Notes

Proposition I.3 is mathematically simple but formally instructive.

In modern numerical language one might describe the theorem as follows: if

$$|AB| > |CD|,$$

then there is a point E on AB such that

$$|AE| = |CD|.$$

The reconstruction deliberately avoids this numerical formulation. "Greater than" is represented by a synthetic segment-order relation, and the conclusion consists of two separate geometric facts:

$$A - E - B \qquad \text{and} \qquad AE \cong CD.$$

This distinction is important. Congruence alone would allow a copy of CD to be placed anywhere; Proposition I.3 requires the copy to occur inside the prescribed longer segment.

The formal theorem therefore combines the two foundational layers that will recur throughout Book I:

$$\text{congruence} \quad + \quad \text{order}.$$

In the Hilbert reconstruction the actual segment-layoff operation is already available at the interface level. Consequently the public Euclidean theorem is relatively short: its main role is to package layoff together with the betweenness condition dictated by the strict segment comparison.

10 Dependency Summary

```
Classical:
I.1 -> I.2 -> circle construction -> I.3

Formal:
HilbertSegmentLess
        |
        +-- witness E
        +-- A-E-B
        +-- CD ~= AE
        |
congruence symmetry
        |
euclid_proposition_3
```

The contrast is especially sharp: the classical proposition performs a construction, whereas the formal theorem exposes existential data already encoded by the Book Zero segment-order relation.

11 Conclusion

Proposition I.3 completes the first elementary segment-construction block of Book I.

Proposition I.2 established free transport of a segment length to a prescribed point. Proposition I.3 adds order: the copied segment is cut from a longer prescribed segment and its endpoint must lie between the original endpoints.

The formalization makes this distinction particularly clear. The theorem is not simply about equality of lengths; it is about the interaction of segment congruence with betweenness.

This combination of congruence and order becomes fundamental throughout the later development. I.3 therefore serves as a compact early example of the two Hilbert layers working together before Book I turns in Proposition I.4 to the rigidity theory of triangles.